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Approximation algorithm for the minimum partial connected Roman dominating set problem

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Abstract

Given a graph $G = (V, E)$ and a function $r : V \mapsto \{0, 1, 2\}$, a node $v \in V$ is said to be *Roman dominated* if $r(v) = 1$ or there exists a node $u \in N_G[v]$ such that $r(u) = 2$, where $N_G[v]$ is the closed neighbor set of v in G. For $i \in \{0, 1, 2\}$, denote V_r^i as the set of nodes with value i under function r. The cost of r is defined to be $c(r) = |V_r^1| + 2|V_r^2|$. Given a positive integer $Q \leq |V|$, the *minimum partial connected Roman dominating set* (MinPCRDS) problem is to compute a minimum cost function r such that at least Q nodes in G are Roman dominated and the subgraph of G induced by $V_r^1 \cup V_r^2$ is connected. In this paper, we give a $(3 \ln |V| + 9)$-approximation algorithm for the MinPCRDS problem.

Keywords Partial connected Roman dominating set · Approximation algorithm · Quota Steiner tree

1 Introduction

The Roman domination problem comes from a consideration for protecting the Roman empire (Stewart 1999). Given a graph $G = (V, E)$, each node represents a location in the Roman Empire. A location is secured by legions stationed there, and a location without a legion is protected by sending armies from neighboring regions. It was decreed by Emperor Constantine the Great that a legion cannot be sent from a secured location to an unsecured location if doing so leaves that location unsecured. Therefore

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only a location with at least two legions can send a legion to its neighboring location. Based on such a background, Cockayne et al. (2004) proposed the model of Roman domination. Given a graph $G = (V, E)$, a function $r : V \mapsto \{0, 1, 2\}$ is called a *Roman dominating function* if for each $v \in V$ with $r(v) = 0$, there exists a node $u \in V$ neighboring v with $r(u) = 2$. For $i \in \{0, 1, 2\}$, denote $V_r^i = \{u \in V : r(u) = i\}$. Define the cost of r as $c(r) = \sum_{v \in V} r(v) = |V_r^1| + 2|V_r^2|$, which is the number of total legions. The *minimum Roman dominating set* (MinRDS) problem is to find a minimum cost Roman dominating function r on G. In the real world, for the security of transmitting information, it is desired that the locations with legions are connected. Such a consideration leads to the *minimum connected Roman dominating set* (MinCRDS) problem, the goal of which is to compute a minimum cost Roman dominating function r such that the subgraph of G induced by $V_r^1 \cup V_r^2$, denoted as $G[V_r^1 \cup V_r^2]$, is connected. The Roman domination problem and its variants also find applications in the field of security in wireless sensor networks (Li et al. 2022; Wang et al. 2020).

In (2000), Dreyer showed that MinRDS is NP-hard in general graphs. Chakradhar and Reddy (2020) showed that MinRDS for star convex bipartite graphs and comb convex bipartite graphs cannot be approximated within factor $(1 - \varepsilon) \ln |V|$ for any $\varepsilon > 0$ unless $NP \subseteq DTIME(|V|^{O(\log \log |V|)})$. They also showed that for general graphs with maximum degree 5, MinRDS is APX-hard. Shang et al. (2010) proved that both MinRDS and MinCRDS are NP-hard even on unit disk graphs.

Note that securing all locations is costly and in some cases it is sufficient to secure some locations. With such a consideration, we propose the *minimum partial connected Roman dominating set* (MinPCRDS) problem. For a graph $G = (V, E)$ and a function $r : V \mapsto \{0, 1, 2\}$, a node $v \in V$ is said to be *Roman dominated under r* if $r(v) = 1$ or there exists a node $u \in N_G[v]$ with $r(u) = 2$, where $N_G[v]$ is the closed neighbor set of v, which contains v itself and all nodes adjacent to v. Given a positive integer $Q \leq |V|$, the goal of MinPCRDS is to compute a function r with the minimum $c(r)$ such that $G[V_r^1 \cup V_r^2]$ is connected and at least Q nodes in G are Roman dominated under r. An instance of MinPCRDS is denoted as $R = (G, Q)$.

Note that MinCRDS is a special case of MinPCRDS when $Q = |V|$. So all the above NP-hardness results for the MinCRDS problem also hold for the MinPCRDS problem. To our knowledge, this paper is the first paper studying MinPCRDS.

1.1 Related work

After Cockayne et al. (2004) have proposed the model of Roman domination, there are many researches on it (Chellali et al. 2021; Dreyer Jr 2000; Haynes et al. 2020), especially on special graphs, such as trees (Henning 2002), interval graphs (Liedloff et al. 2005) and unit disk graphs (Shang et al. 2010). There are also many variants of Roman domination, such as *maximal Roman domination* (Ahangar et al. 2017), *total Roman reinforcement* (Ahangar et al. 2019), *triple Roman domination* (Ahangar et al. 2021), *double Roman domination* (Beeler et al. 2016) and *Italian domination* (Chellali et al. 2016). Many of these studies are from graph theoretical point of view or designing polynomial-time algorithms for special cases. In the following, we only focus on results related with approximation algorithms.

For MinRDS in general graphs, Chakradhar and Reddy (2020) proposed a $(2 + 2\ln(\Delta + 1))$-approximation algorithm, where Δ is the maximum degree of graph G. Recently, Li et al. (2022) gave a greedy algorithm for MinRDS in general graphs which improved the approximation ratio to $H(\Delta + 1)$, where $H(\cdot)$ is the Harmonic number. Note that this ratio has aligned with the lower bound of MinRDS. For MinCRDS in general graphs, they gave a greedy algorithm with approximation ratio of $(1 + \varepsilon)\ln\Delta + O(1)$, where $\varepsilon > 0$ is a constant. This ratio is asymptotically tight for MinCRDS. Shang et al. (2010) presented approximation algorithms for MinRDS and MinCRDS on unit disk graphs, achieving approximation ratios of 5 and 7, respectively.

For researches on variants of Roman domination, Chakradhar and Reddy gave a $3(\ln(\Delta - 0.5) + 1.5)$-approximation algorithm for the *minimum total Roman 3-dominating function problem* in Chakradhar and Reddy (2021) and a $2(\ln(\Delta - 0.5) + 1.5)$-approximation algorithm for the *minimum total Roman 2-dominating function problem* in Chakradhar and Reddy (2022). Wang et al. (2020) gave a $6(k + 2)$-approximation algorithm for the *minimum connected strong k-Roman dominating set problem* in unit ball graphs. Li and Zhang (2023) gave a greedy algorithm for the *minimum Italian dominating function problem*, achieving approximation ratio of $2 + \ln(\frac{\Delta}{2} + 1)$. They also gave a greedy algorithm for the *minimum connected Italian-dominating function problem*, achieving approximation ratio of $4 + \ln(\frac{\Delta}{2} - 1)$.

1.2 Our contribution

In this paper, we show that making use of a β-approximation algorithm for the *minimum quota node weighted Steiner tree* (QNWST) problem, we can obtain a 3β-approximate solution to the MinPCRDS problem. Note that Moss and Yuval have dessigned a $(\log|V| + 3)$-approximation algorithm for QNWST in Moss and Rabani (2007). So, MinPCRDS has a $(3\log|V| + 9)$-approximate solution.

Observe that making use of an α-approximation algorithm for the *minimum partial connected dominating set* (PCDS) problem, it is not hard to obtain a 2α-approximate solution to the MinPCRDS problem. Since Khuller et al. (2020) have designed a $(4\log|V| + 2 + o(1))$-approximation algorithm for MinPCDS, MinPCRDS has a $(8\log|V| + 2 + o(1))$-approximate solution. Our algorithm outperforms this ratio by a factor of $\frac{3}{8}$.

The remaining part of this paper is organized as follows. Section 2 is the main part. After introducing some fundamental notations and terminologies in Sect. 2.1, the algorithm is presented in Sect. 2.2, its theoretical analysis is given in Sect. 2.3, and Sect. 2.4 reveals a relation between MinPCDS and MinPCRDS. Section 3 concludes the paper and discusses some directions for future work.

2 The algorithm and its analysis

2.1 Preliminaries

Given a graph $G = (V, E)$, the *neighborhood* of node $v \in V$, denoted as $N_G(v)$, is the set of nodes adjacent to v in G. The *closed neighborhood* of node v is $N_G[v] = N_G(v) \cup \{v\}$. The closed neighborhood of node set $S \subseteq V$ is $N_G[S] = \bigcup_{v \in S} N_G[v]$. A node v is *dominated by a node set* S if either $v \in S$ or v has a neighbor in S. Note that the set of nodes dominated by S is $N_G[S]$. Denote by $G[S]$ the subgraph of G induced by $S \subseteq V$, which is the graph on node set S and contains all those edges of E whose two ends both belong to S. For a positive integer $Q \leq |V|$, the *minimum partial connected dominating set* (MinPCDS) problem is to find a smallest node subset $S \subseteq V$ such that $|N_G[S]| \geq Q$ and $G[S]$ is connected.

For a function $r : V \mapsto \{0, 1, 2\}$, when a node v is Roman dominated under r, we also say that v is *Roman dominated by node set* $S = V_r^1 \cup V_r^2$, and denote the set of nodes Roman dominated by S as $D_r(S)$. Clearly, $D_r(S) \subseteq N_G[S]$. Using these notations, the MinPCRDS problem is to find a minimum cost function $r : V \mapsto \{0, 1, 2\}$ such that the node set $S = V_r^1 \cup V_r^2$ has $|D_r(S)| \geq Q$ and $G[S]$ is connected.

Our algorithm will make use of an approximation algorithm for the *minimum quota node weighted Steiner tree* (MinQNWST) problem. Given a graph $G' = (V', E')$, a quota Q', a non-negative profit function $p' : V' \mapsto \mathbb{R}^+$ and a nonnegative cost function $c' : V \mapsto \mathbb{R}^+$, the goal of MinQNWST is to find a subtree T' of G' such that the cost of T', which is $c'(T') = \sum_{v \in V'(T')} c'(v)$, is minimized subject to the constraint that the profit of T', which is $p'(T') = \sum_{v \in V'(T')} p'(v)$, is at least Q'. An instance of MinQNWST is denoted as $I' = (G', c', p', Q')$.

2.2 The algorithm

Given a MinPCRDS instance $R = (G, Q)$, we construct a MinQNWST instance $I' = (G', c', p', Q')$ as follows. The node set of G' is $V' = A \cup B$, where $A = \{v \colon v \in V\}$ and $B = \{D_v \colon v \in V\}$. That is, every node $v \in V$ corresponds to a node $D_v \in B$. Call nodes in A as A-*nodes* and nodes in B as B-*nodes*. There is no edge between A-nodes. An A-node v is adjacent to every B-node D_u with $u \in N_G[v]$. For each pair of B-nodes D_u, D_v with $N_G[u] \cap N_G[v] \neq \emptyset$, link them by an edge in G'. Let $Q' = Q$. Define cost function c' on V' as

$$\begin{cases} c'(v) = 0 \text{ for } v \in A, \\ c'(D_v) = 1 \text{ for } D_v \in B. \end{cases} \tag{1}$$

Define profit function p' on V' as

$$\begin{cases} p'(v) = 1 \text{ for } v \in A, \\ p'(D_v) = 0 \text{ for } D_v \in B. \end{cases} \tag{2}$$

The construction is illustrated by Fig. 1.

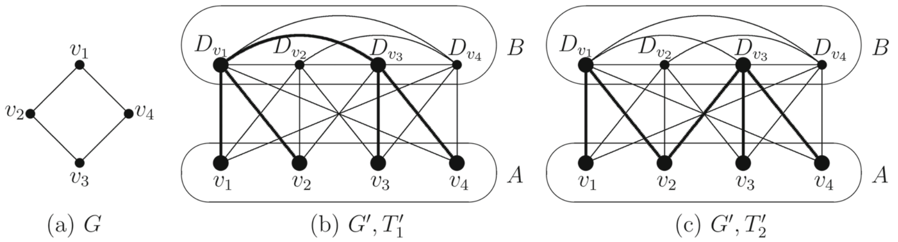

Fig. 1 (*a*) and (*b*) illustrate the construction of graph G'. (*b*) and (*c*) illustrate the following property (3)

The algorithm first finds a subtree tree T' for the MinQNWST instance I' constructed as the above. Let $A(T')$ and $B(T')$ be the set of A-nodes and the set of B-nodes on T', respectively. We may assume that

$$\text{no } B\text{-nodes are adjacent on } T'. \tag{3}$$

In fact, if there are two B-nodes $D_{v_1}, D_{v_2} \in B(T')$ that are adjacent on T', then by the construction of G', we have $N_G[v_1] \cap N_G[v_2] \neq \emptyset$. Let v be a node in $N_G[v_1] \cap N_G[v_2]$. Replacing edge $D_{v_1}D_{v_2}$ by path $D_{v_1}vD_{v_2}$ will not decrease the profit of the tree while the cost is kept the same (because of the construction of c' and p'). For example, consider the example in Fig. 1. Tree T_1' indicated by the blackened edges and bigger nodes in (*b*) contains edge $D_{v_1}D_{v_3}$. Replacing edge $D_{v_1}D_{v_3}$ by the path $D_{v_1}v_2D_{v_3}$, one obtains tree T_2' as indicated in (*c*). Trees T_1' and T_2' have the same cost and the the same profit.

After constructing a tree T' satisfying property (3), the algorithm picks all those A-nodes corresponding to the B-nodes on T' into the solution. Combining (3) with the fact that A-nodes are independent, any pair of B-nodes $D_{v_1}, D_{v_2} \in B(T')$ that are nearest on T' are linked by a path of T' with the form $D_{v_1}vD_{v_2}$, where v is an A-node in $N_G[v_1] \cap N_G[v_2]$. For such a pair D_{v_1} and D_{v_2}, if the corresponding A-nodes v_1, v_2 are non-adjacent in G, then node v is added into the solution (see line 4 of the algorithm).

Algorithm 1 MinPCRDS(G, Q)

Input: A graph $G = (V, E)$ and a positive integer $Q \leq |V|$
Output: A function $r : V \mapsto \{0, 1, 2\}$.

1: construct a MinQNWST instance $I' = (G', c', p', Q')$ and compute a subtree T' satisfying (3) using a β-approximation algorithm for MinQNWST;
2: $S \leftarrow \{v : D_v \in B(T')\}$; $S' \leftarrow \emptyset$;
3: **for** each pair of nodes D_{v_1}, D_{v_2} that are linked on T' by a path of form $D_{v_1}vD_{v_2}$ **do**
4: if v_1, v_2 are not adjacent in G, then $S' \leftarrow S' \cup \{v\}$;
5: **end for**
6: **return** node function $r(v) = \begin{cases} 2, & v \in S \\ 1, & v \in S' \\ 0, & v \in V \backslash (S \cup S'). \end{cases}$

2.3 Theoretical analysis

We first prove the correctness of the algorithm.

Theorem 2.1 *The function r computed by Algorithm 1 is a feasible solution to the MinPCRDS instance $R = (G, Q)$.*

Proof Note that $V_r^1 = S$ and $V_r^2 = S'$. We first show that $G[S \cup S']$ is connected. If $G[S]$ is connected, then $G[S \cup S']$ is also connected, because the nodes added in line 4 of the algorithm are adjacent to nodes in S. Next, suppose $G[S]$ is not connected. Because T' is a tree connecting all those B-nodes corresponding to the A-nodes of S, there must exist a pair of B-nodes $D_{v_1}, D_{v_2} \in B(T')$ that are linked by a path $D_{v_1} v D_{v_2}$ on T' with $v \in N_G[v_1] \cap N_G[v_2]$ and v_1, v_2 belong to different connected components of $G[S]$. Then line 4 of the algorithm adds node v into S', which reduces the number of connected components by at least one. Iteratively carrying out such an argument, at the end of the for loop, graph $G[S \cup S']$ is connected.

Next, we show that at least Q nodes are Roman dominated by $S \cup S'$. Since no nodes in A are adjacent and T' is connected, any A-node $u \in A(T')$ is linked by an edge of T' to a B-node $D_v \in B(T')$. Then, by the construction of graph G' and the definition of S, we have $A(T') \subseteq N_G[S]$. Combining this observation with the definition of p' in (2) and because T' is a feasible solution to the MinQNWST instance I', we have

$$Q' \leq p'(T') = \sum_{u \in A(T')} p'(u) = |A(T')| \leq |N_G[S]|. \tag{4}$$

Note that $Q = Q'$. Also note that for any node $v \in S$, since $r(v) = 2$, any node $u \in N_G[v]$ is Roman dominated by v. So, (4) implies that at least Q nodes are Roman dominated by S, and thus also by $S \cup S'$. □

The next lemma lower bounds the optimal value of a MinPCRDS instance in terms of the optimal value of its corresponding MinQNWST instance.

Lemma 2.2 *Denote by opt_{RDS} and opt_{ST} the optimal values of a MinPCRDS instance $R = (G, Q)$ and the optimal value of its corresponding MinQNWST instance $I' = (G', c', p', Q')$, respectively. Then $opt_{ST} \leq opt_{RDS}$.*

Proof Suppose r^* is an optimal solution to the MinPCRDS instance R. Let $S^* = V_{r^*}^1 \cup V_{r^*}^2$. Then $G[S^*]$ is connected and $|D_{r*}(S^*)|$, the number of nodes that are Roman dominated by S^*, is at least Q. By noting that $D_{r*}(S^*) \subseteq N_G[S^*]$, we have

$$|N_G[S^*]| \geq Q. \tag{5}$$

By the construction of G', for any pair of A-nodes v_1, v_2 that are adjacent in G, their corresponding B-nodes D_{v_1} and D_{v_2} are adjacent in G'. So, the induced subgraph $G'[B(S^*)]$ is connected, where $B(S^*) = \{D_v : v \in S^*\}$. Note that every A-node $u \in N_G[S^*]$ is linked to a B-node of $B(S^*)$. So, the induced subgraph $G'[B(S^*) \cup N_G[S^*]]$

is connected and thus has a spanning tree T'. Then by (5) and the definition of p',

$$p'(T') = \sum_{u \in N_G[S^*]} p'(u) = |N_G[S^*]| \geq Q.$$

This implies that T' is a feasible solution to the MinQNWST instance I'. So,

$$opt_{ST} \leq c'(T'). \tag{6}$$

By the definition of c' in (1), we have

$$c'(T') = \sum_{D_v \in B(S^*)} c'(D_v) = |B(S^*)| = |S^*| = |V^1_{r^*}| + |V^2_{r^*}|. \tag{7}$$

Note that

$$opt_{RDS} = |V^1_{r^*}| + 2|V^2_{r^*}| \geq |V^1_{r^*}| + |V^2_{r^*}|. \tag{8}$$

Then $opt_{ST} \leq opt_{RDS}$ follows from inequalities (6) to (8). □

Now, we are ready to analyze the approximation ratio of the algorithm.

Theorem 2.3 *Algorithm 1 is a* 3β*-approximation for the MinPCRDS problem.*

Proof Since T' is a β-approximate solution to the MinQNWST instance I', we have

$$c'(T') \leq \beta \cdot opt_{ST}. \tag{9}$$

By the construction of S in line 2 of the algorithm and the definition of c',

$$c'(T') = \sum_{D_v \in B(T')} c'(D_v) = |B(T')| = |S|. \tag{10}$$

Because T' is a tree, there are at most $|B(T')| - 1 = |S| - 1$ pairs of B-nodes satisfying the condition of the for loop of Algorithm 1. So

$$|S'| \leq |S| - 1. \tag{11}$$

Combining inequalities (9) to (11), Lemma 2.2 and the observation that $|S| = |V_r^2|$ and $|S'| = |V_r^1|$, the cost of function r has

$$c(r) = |V_r^1| + 2 \cdot |V_r^2| < 3|S| = 3c'(T') \leq 3\beta \cdot opt_{ST} \leq 3\beta \cdot opt_{RDS}.$$

The approximation ratio 3β is proved. □

In (2007), Moss and Rabani gave a $(3 + \log |V|)$-approximation algorithm for the MinQNWST problem. As a consequence, we have the following result.

Corollary 2.4 *There is a* $(3 \log |V| + 9)$*-approximation algorithm for MinPCRDS.*

2.4 A relation between MinPCDS and MinPCRDS

Suppose $\mathcal{A}_{PCDS}$ is an α-approximation algorithm for the MinPCDS problem. Making use of $\mathcal{A}_{PCDS}$, we can find a 2α-approximate solution to a MinPCRDS instance $R = (G, Q)$ as follows: view (G, Q) as a MinPCDS instance; use $\mathcal{A}_{PCDS}$ on (G, Q) to find a node set S with $|N_G[S]| \geq Q$ and $G[S]$ being connected; set

$$r(v) = \begin{cases} 2, & v \in S, \\ 0, & v \in V \backslash S. \end{cases}$$

Then $V_r^1 = \emptyset$ and $V_r^2 = S$. All those nodes in $N_G[S]$ are Roman dominated by S. So, $|D_r(S)| \geq Q$ and thus r is a feasible solution to the MinPCRDS instance $R = (G, Q)$. Since $c(r) = 2|S| \leq 2\alpha opt_{PCDS}$ and $opt_{PCDS} \leq opt_{PCRDS}$, where opt_{PCDS} and opt_{PCRDS} are the optimal values of the MinPCDS instance and the MinPCRDS instance, respectively, approximation ratio 2α follows.

In (2020), Khuller et al. designed a $(4 \ln |V| + 2 + o(1))$-approximation algorithm for the MinPCDS problem. So, MinPCRDS admits an $(8 \ln |V| + 4 + o(1))$-approximation. Note that Algorithm 1 has approximation ratio at most $(3 \log |V| + 9)$ (see Corollary 2.4), which is only $3/8$ of the ratio obtained by this naive method.

3 Conclusion and future work

In this paper, we prove that a β-approximation algorithm for the MinQNWST problem can yield a 3β-approximate solution to the MinPCRDS problem. Making use of a $(3 + \log |V|)$-approximation algorithm for the MinQNWST problem in Moss and Rabani (2007), the MinPCRDS problem can be approximated within factor $(3 \log |V| + 9)$.

The crucial idea of our method is how to transform the current problem into a node-weighted Steiner tree problem. This method can also be used on the *partial k-Roman connected dominating set* (PkRCDS) problem and the *budgeted Roman connected dominating set* (BRCDS) problem. However, a naive implementation of this method cannot yield satisfactory approximation ratios. In fact, for the PkRCDS problem, parameter k appears in the approximation ratio, which might not be good if k is large. How to remove k from the approximation ratio needs further explorations. For the BRCDS problem, making use of an algorithm for the *budgeted node-weighted Steiner tree* (BNWST) problem, our methods yields an $O(1/\log |V|)$-approximation. The crux lies in the hardness of the BNWST problem. Note that the BNWST instance resulted from the transformation has some specialities. It is interested to ask whether such a special BNWST problem has a constant approximation.

Note that our method cannot be generalized to yield a good approximation for the cost version. Given a graph $G = (V, E)$, a non-negative cost function $c : V \mapsto \mathbb{R}^+$ (stationing legions at different locations may incur different costs), a weight function $w : V \mapsto \mathbb{R}^+$ (securing different locations may have different values to the nation), and a non-negative number $Q \leq \sum_{v \in V} w(v)$, the goal of the *minimum cost partial connected Roman dominating set* (MinCPCRDS) problem is to compute a function $r : V \mapsto \{0, 1, 2\}$ with the minimum cost $c(r) = \sum_{v \in V_r^1} c(v) + 2 \sum_{v \in V_r^2} c(v)$ such that $G[V_r^1 \cup V_r^2]$ is connected and the weight of those Roman dominated nodes under r, which is $w(r) = \sum_{v \in D_r(V_r^1 \cup V_r^2)} w(v)$, is at least Q. Note that in Algorithm 1, to ensure the connectivity of the output, we add S' to connect components of $G[S]$. For the cardinality version, $|S'|$ can be upper bounded by $|S| - 1$. However, for the cost version, it is hard to upper bound $c(S')$. New ideas have to be further explored to solve the MinCPCRDS problem.

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Declarations

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